MODULE 02 · DOMAIN 1 - SOLUTION DESIGN AND ARCHITECTURE

From Business Problem to Claude Solution

Where Claude belongs in a system, where it does not, and how to frame the problem before choosing anything.

DOMAIN 1 of 7 EXAM WEIGHT 17% MODULE 2 of 38

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The first design decision is not which model. It is whether Claude **belongs in this step at all.**

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Give Claude the step, or keep it out?

GIVE CLAUDE THE STEP WHEN

Meaning and messiness

- Inputs are varied or unstructured
- The rule is about meaning, not format
- Output is a draft that gets checked
- Categories change often
- A reason must come with the answer

KEEP IT OUT WHEN

Exactness and speed

- The answer is exact arithmetic
- Fixed rules must apply identically
- The budget is milliseconds
- A purpose-built tool already does it
- Nobody would check a binding decision

THE TIEBREAKER

Meaning and messiness go to Claude. Exactness, speed and fixed rules stay in code.

The right component for each job

THE JOB	BEST COMPONENT
Totals, tax, interest, pro-rata refunds	Deterministic code
Score a card payment in 30 ms	A trained fraud model
Find passages across millions of documents	A search or vector index
Read varied documents into fields	Claude, with a schema
Draft a reply a person will approve	Claude
Route tickets as categories keep changing	Claude, zero-shot

ON THE EXAM

Claude is usually one component in a pipeline, not the whole pipeline.

Claude or a trained classifier?

A TRAINED MODEL WINS WHEN

Stable, labelled, fast

- Plenty of labelled examples
- Categories fixed for years
- Huge volume at millisecond latency
- No explanation needed

CLAUDE WINS WHEN

Changing, messy, explainable

- Little labelled data
- Categories change or grow
- Messy text, many languages
- Reasons must come with labels

THE TIEBREAKER

Stable, labelled and fast favours a trained model. Changing, messy and explainable favours Claude.

Frame the problem before choosing anything

01
The decision
What task or decision gets better?

02
The user
Who acts on the output?

03
Today
How is it done now, at what cost?

04
The measure
Which metric proves success?

05
Constraints
Rules, data, latency, budget.

06
Then shape
Where Claude fits, if anywhere.

ON THE EXAM

A solution named before the problem is framed is a guess. Frame first, then design.

Four questions that place Claude in a pipeline

- **What goes in?** Messy and varied favours Claude; clean and structured may not need it.
- **What comes out?** A label, fields, a draft, or an action.
- **Who or what checks it?** Code, a person, both, or nobody.
- **What if it is wrong?** Reversible and cheap, or binding and costly.

> ON THE EXAM
>
> The riskier the output, the more checking around it, and the less Claude should own alone.

One process, many owners

Mortgage applications, from documents to decision.

STEP	OWNER
Receive payslips and statements	Document system
Read them into income and outgoings fields	Claude, with a fixed schema
Check totals, dates and formats	Validation code
Calculate affordability	Rules engine
Decide	A qualified underwriter
Draft the decline letter	Claude, approved by the underwriter

ON THE EXAM

Claude appears twice in this process, and decides nothing.

Find the simplest solution possible, and only increase complexity **when needed**.

The ladder of complexity

RUNG	CLIMB TO IT WHEN
No model	Rules or code already do the job
One call	A good prompt meets the bar
One call + retrieval	It needs your documents or examples
Workflow	Several fixed steps, known in advance
Agent	The steps cannot be predicted
Multi-agent	Big, parallel work that justifies the cost

ON THE EXAM

Each rung adds cost, latency and new ways to fail. Climb only on evidence.

Who makes the decision?

CLAUDE DECIDES

Automation

- Low stakes and reversible
- Errors are cheap to fix
- Checks run on every output
- Example: routing a ticket

CLAUDE PREPARES, A PERSON DECIDES

Augmentation

- Credit, health, jobs, legal status
- Irreversible or binding
- Regulator expects human review
- Example: a lending decision

THE TIEBREAKER

High-risk or irreversible: Claude prepares the decision and a qualified person makes it.

Hard constraints that reshape a design

- **Regulation:** high-risk use needs human review; health data needs covered services.
- **Data boundary:** where processing may happen decides the platform.
- **Latency:** a 50 ms budget removes a model call from that step.
- **Cost ceiling:** per-transaction limits push towards lighter models or no model.
- **Audit:** if a figure must be reproduced exactly, it comes from code.

> ON THE EXAM
>
> Apply these first. Each can move Claude to another step, another platform, or out entirely.

A card payments company blocks fraudulent transactions at the moment of authorisation, within a 30 ms budget per transaction. It has five years of labelled fraud data and a trained scoring model.

Investigators then spend about 40 minutes writing up each flagged case from logs, notes and customer messages.

What should the architect recommend?

A Keep the trained model for scoring, and use Claude to draft investigation write-ups for review

B Replace the scoring model with Claude's fastest tier, since it is cheap and quick

C Score each transaction with Claude at high effort, to catch more subtle fraud

D Score transactions with Claude in an overnight batch, to reduce cost

◆ PAUSE AND DECIDE

What should the architect recommend?

A Keep the trained model for scoring, and use Claude to draft investigation write-ups for review

Each job goes to its best component: fast labelled scoring, messy narrative drafting.

B Replace the scoring model with Claude's fastest tier, since it is cheap and quick

Fast for a model, but no hosted call fits 30 ms, and it wastes five years of labels.

C Score each transaction with Claude at high effort, to catch more subtle fraud

Slower still, in the one step where milliseconds are the constraint.

D Score transactions with Claude in an overnight batch, to reduce cost

Fraud must be stopped at authorisation. Overnight is too late.

THE SIGNAL

A millisecond budget and years of labels: keep the trained model. Messy write-ups: Claude.

> Tallyhawk's support team routes about 1,800 tickets a day into 22 queues. Product teams add or rename features most months, so queues change too. There are only a few hundred labelled tickets, and customers write in six languages.
>
> Team leads want a short reason with every routing decision, so they can spot mistakes.

Which approach fits best?

A. Claude classifies each ticket using queue descriptions in the prompt, returning a queue and a short reason

B. Train a custom classifier on last year's labelled tickets and retrain it every quarter

C. Keyword rules maintained by team leads, translated into each of the six languages

D. A Claude agent with tools to investigate each customer's account before routing

◆ PAUSE AND DECIDE

Which approach fits best?

A Claude classifies each ticket using queue descriptions in the prompt, returning a queue and a short reason

Changing queues, little data, many languages, reasons wanted: every signal for Claude.

B Train a custom classifier on last year's labelled tickets and retrain it every quarter

A few hundred labels, and queues that change monthly, defeat quarterly retraining.

C Keyword rules maintained by team leads, translated into each of the six languages

Brittle on meaning, six rule sets to maintain, and no reasons.

D A Claude agent with tools to investigate each customer's account before routing

Over-engineered. Routing is one classification, not an open-ended task.

THE SIGNAL

Changing categories, little labelled data, many languages, reasons required: one Claude call.

> Quillon Health wants staff to ask questions about its 220-page employee benefits handbook. The handbook changes twice a year and holds no patient data.
>
> One engineer proposes a full retrieval pipeline with chunking, embeddings and a vector database. Another proposes a research agent that searches the handbook with tools.

What is the simplest design likely to meet the need?

A. A retrieval pipeline with chunking, embeddings and a vector database

B. A research agent that searches the handbook with tools on every question

C. Put the whole handbook in the prompt as a cached prefix, and ask Claude to cite sections

D. Several specialist agents, one for each chapter, coordinated by a lead agent

◆ PAUSE AND DECIDE

What is the simplest design likely to meet the need?

A A retrieval pipeline with chunking, embeddings and a vector database

Sound at scale, but a whole pipeline to build and run for one small document.

B A research agent that searches the handbook with tools on every question

Adds steps, latency and cost to what a single call can answer.

C Put the whole handbook in the prompt as a cached prefix, and ask Claude to cite sections

Small and stable: one call sees everything, and caching cuts repeat cost.

D Several specialist agents, one for each chapter, coordinated by a lead agent

The top of the ladder for a question-and-answer job on one book.

THE SIGNAL

Under about 200,000 tokens and stable: put it in the prompt and cache it. RAG is for bigger corpora.

A water company texts each customer their exact bill amount and due date when a bill is issued. The billing system already holds both figures. About 900,000 messages go out each month.

A product manager proposes using Claude to write each text, so the messages feel more personal.

What should the architect recommend?

A. Claude writes each text from the account record, using the fastest tier to keep costs down

B. A fixed, approved template that code fills with the amount and date

C. Claude writes each text, with a structured output schema to guarantee the figures

D. Claude writes each text, and a sample of messages is reviewed weekly

◆ PAUSE AND DECIDE

What should the architect recommend?

A. Claude writes each text from the account record, using the fastest tier to keep costs down

Cheaper per message, but still a model generating figures that code already has.

B. A fixed, approved template that code fills with the amount and date

Exact figures, identical wording, near-zero cost. Nothing here needs a model.

C. Claude writes each text, with a structured output schema to guarantee the figures

A schema fixes the shape, not whether the amount is right. Code already has it.

D. Claude writes each text, and a sample of messages is reviewed weekly

Sampling finds errors after customers receive them. The template prevents them.

THE SIGNAL

Fixed content, exact figures already in a system, huge volume: no model. A template.

> Quillon's chief medical officer asks Corvane for "an AI agent for prior authorisations" and wants a demonstration next month.
>
> Nobody has yet said which part of the process is slow, who would use the agent's output, or how success would be measured.

What should the architect do first?

A Choose between the Opus and Sonnet tiers, so costs can be estimated

B Build a quick prototype agent, so the officer can react to something real

C Select an agent framework and hosting platform for the build

D Establish the decision to improve, who uses the output, today's baseline and the success measure

◆ PAUSE AND DECIDE

What should the architect do first?

A Choose between the Opus and Sonnet tiers, so costs can be estimated

Model choice follows requirements. There are none yet.

B Build a quick prototype agent, so the officer can react to something real

A prototype of an unframed problem tests the wrong thing, quickly.

C Select an agent framework and hosting platform for the build

Assumes an agent is the answer before anyone knows the question.

D Establish the decision to improve, who uses the output, today's baseline and the success measure

Framing comes first. Every later choice depends on it.

THE SIGNAL

A solution named before the problem is framed: frame first, whatever the deadline.

> Tallyhawk refunds part of a month's fee when a customer downgrades their plan mid-month. Customers request downgrades in free-text emails, in many languages, and support wants a plain-English explanation sent with each refund.
>
> Finance must be able to reproduce any refund figure exactly in an audit, and the figure is shown in checkout within a 50 ms budget.

Which two details keep the refund calculation in deterministic code? Choose two.

- A Requests arrive as free-text emails
- B Customers write in many languages
- C Finance must reproduce every figure exactly in an audit
- D Support wants a plain-English explanation with each refund
- E Checkout shows the figure within a fifty millisecond budget

◆ PAUSE AND DECIDE

Which two details keep the refund calculation in deterministic code? Choose two.

A. Requests arrive as free-text emails

A reason to use Claude for reading requests, a different step.

B. Customers write in many languages

Another point for Claude on the way in, not for the calculation.

C. Finance must reproduce every figure exactly in an audit

Exact reproducibility means tested code, every time.

D. Support wants a plain-English explanation with each refund

A drafting job for Claude, using the figure code produced.

E. Checkout shows the figure within a fifty millisecond budget

No model call fits that budget. Code does.

THE SIGNAL

One workflow, several components: Claude reads and explains, code calculates.

In five lines

- **First ask whether Claude belongs in the step**, then which model.
- **Meaning and messiness** to Claude; **exactness, speed and fixed rules** to code.
- **Frame the problem first:** decision, user, baseline, measure, constraints.
- **Climb the ladder only on evidence.** Small and stable? Put it in the prompt and cache it.
- **High-risk decisions:** Claude prepares, a qualified person decides.

> COMING UP
>
> Next: the patterns themselves. Augmented LLM, five workflows, and when an agent earns its cost.